How-To Library

Informatica

Set up OAuth with Azure AD

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Abstract

If your organization is licensed and configured for SAML, you can use OAuth to start a REST API session using JSON web token (JWT) access tokens. This article explains how to configure OAuth using Azure Active Directory (Azure AD) as the identity provider.

Supported Versions

- Informatica Intelligent Cloud Services December 2023

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Overview

If you are an Informatica Intelligent Cloud Services administrator and your organization is licensed and configured to use SAML, you can set up OAuth using Azure AD as the identity provider so that users can start REST API sessions using JWT access tokens.

Using JWT access tokens to log in to Informatica Intelligent Cloud Services is similar to using SAML assertions. However, unlike using SAML assertions, users must retrieve the JWT access tokens to use in login requests.

Before your organization can use JWT access tokens, complete the following tasks:

1. Configure the organization to use SAML for authentication, and set up users as SAML users.
2. Register an OAuth application. This article describes the steps to register an application with Azure AD as the identity provider.
3. Set up a method to retrieve JWT access tokens from the identity provider.
4. Register the identity provider using the IdentityProviders resource in the Informatica Intelligent Cloud Services REST API.

To log in, users obtain a JWT access token from the identity provider and include the token in a loginOAuth POST request. The token can be used for one REST API session. If the login request is successful, the response includes a session ID to use in subsequent API calls.

This article describes how to create an OAuth application in Azure AD. For more information about using Azure AD, see the Azure AD documentation.

For information about SAML configuration, see SAML single sign-on.

For information about registering an identity provider and logging in to Informatica Intelligent Cloud Services using JWT access tokens, see the following topics in *REST API Reference*:

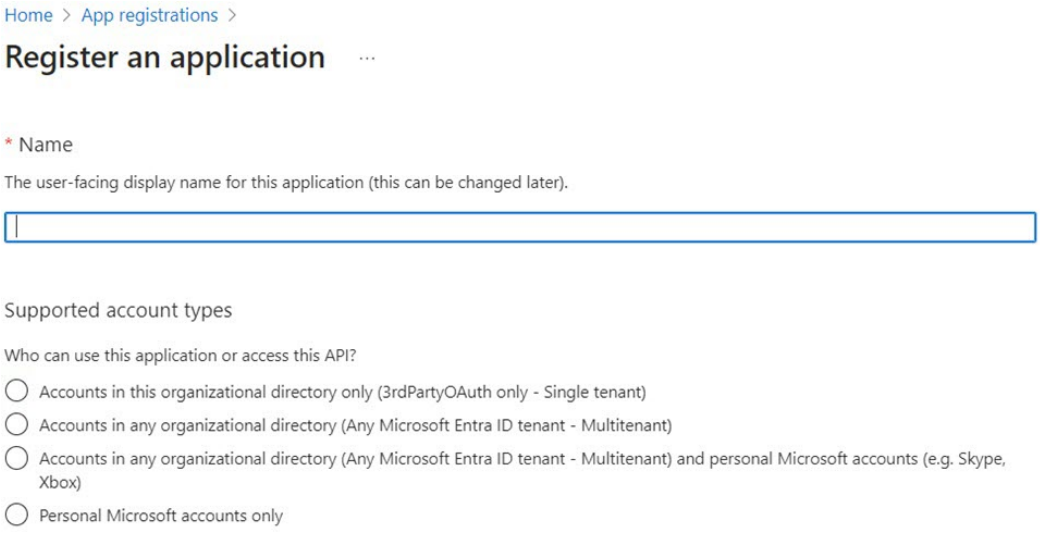
- Identity providers
- loginOAuth

Step 1. Register an application

Register a client application to use for Oauth on the **App registrations** page.

1. Log in to Azure AD and switch to the tenant where your SAML application is configured.
2. Select **App registrations > New registration**.

Figure 1.



3. Create a display name for the application.
4. Enter the account type.
5. Enter a redirect URI that points to the application.
6. Click **Register**.

Step 2. Generate the client secret

Generate a client secret for the application. The client secret is the password for the application and is used in POST requests to fetch a token.

1. Select **Overview > Client credentials > New client secret**.
2. Create a client secret for the application.

Step 3. Define the application URI

Define an application URI that points to your Informatica Intelligent Cloud Services organization.

1. Select **Overview > Application URI**.

2. Enter the URI for your application using the following format: api://<organization ID>.<cloud provider>-<region>.informatica.com

   For example:

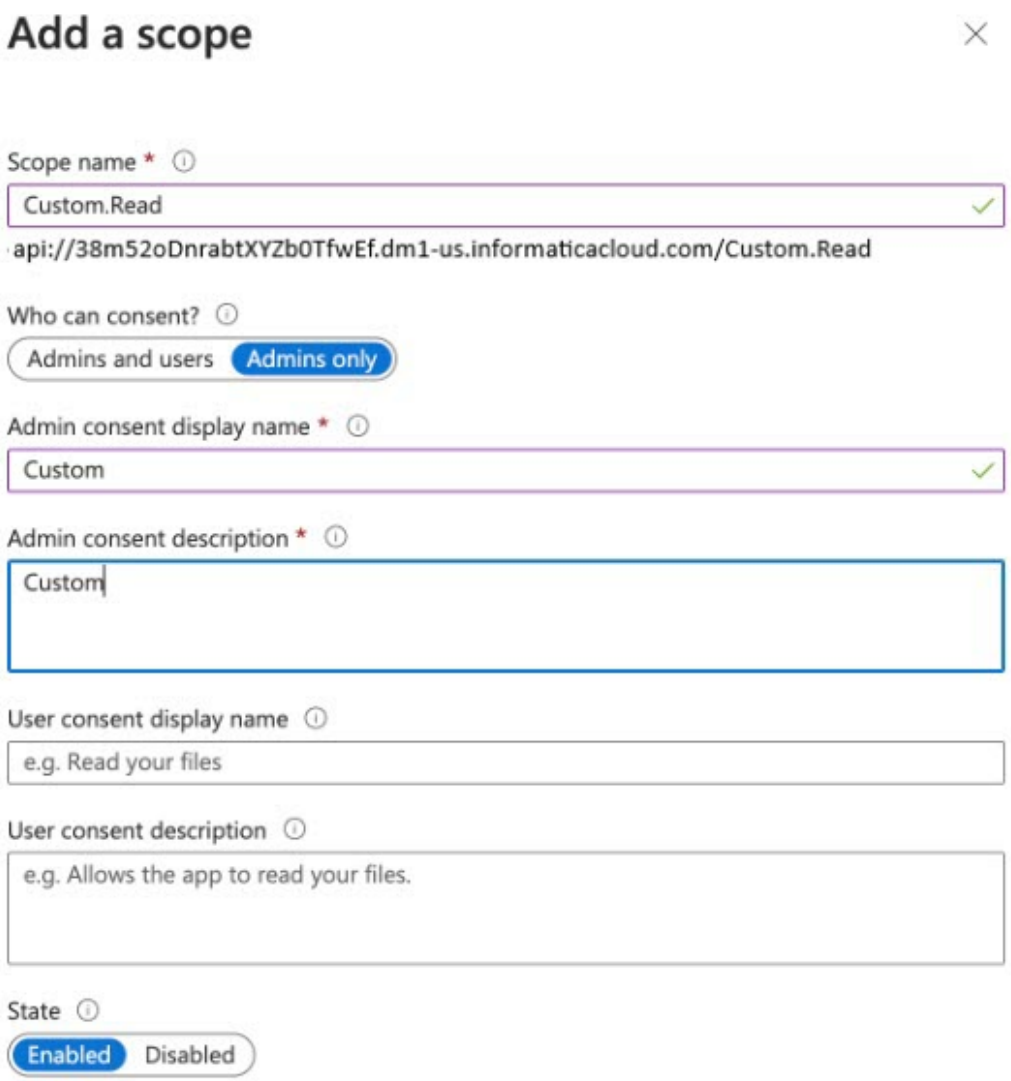
   ```
   api://38m52oDnrabtXYZb0TfEf.dm1-us.informaticacloud.com
   ```

Step 4. Configure API permissions

Configure API permissions for the client application.

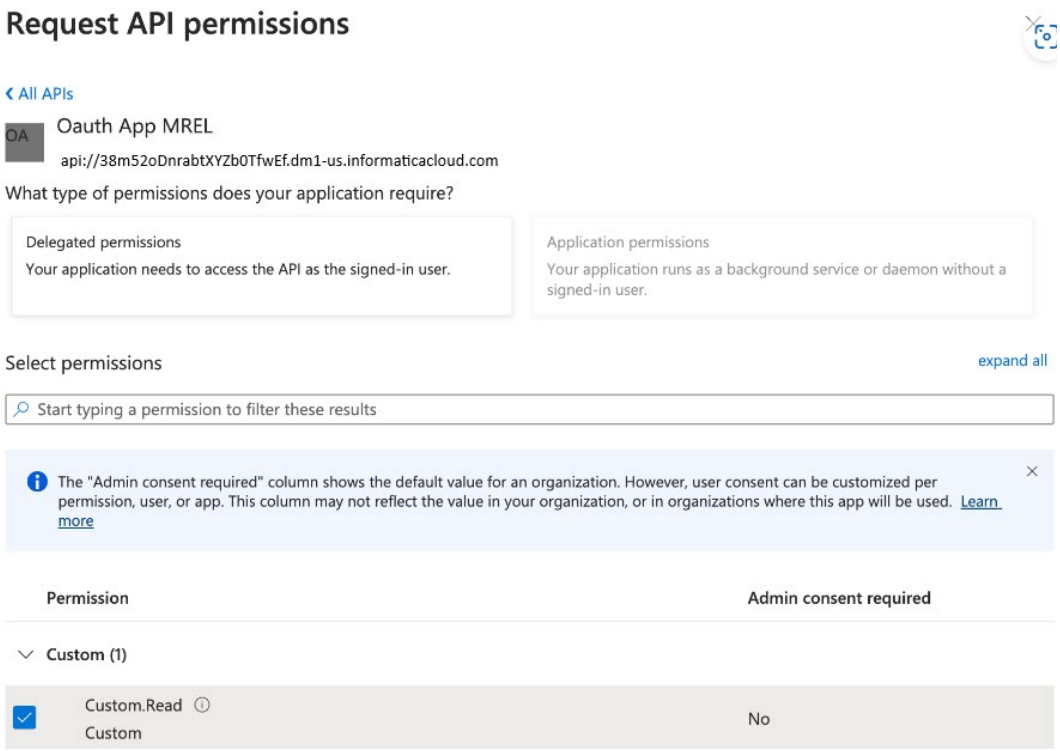
1. Select **Expose an API**.
2. Define a scope for the API. For example, `custom.read`.

Figure 2.



3. Click **Add a client application** and search for the client application using the client ID that was created when you registered the application. Select the application and click **Add application**.
4. Select **API permissions > Add a permission > APIs my organization uses** and select the application name.
5. Select **Delegated permissions** and under **Select permissions**, select the scope that you defined.

Figure 3.



6. Select **Authentication** and add the following URI: `https://jwt.ms`

Step 5. Configure the token

Configure the token type.

1. Select **Token configuration > Access tokens**.
2. For the supported account type, select **Accounts in this organizational directory only**.
3. Under **Advanced settings**, allow public client flows.

Fetching tokens

To fetch tokens, use one of the following methods:

- If multi-factor authentication is set up for the tenant, use user credentials.
- If multi-factor authentication isn't set up for the tenant, use authorization codes.

Fetch tokens using user credentials

You can fetch a token using your Azure AD user name and password.

Send a POST request using the following URI:

```
https://login.microsoftonline.com/<tenant ID>/oauth2/v2.0/token
```

Include the following fields in the request:

Field	Description
grant_type	Use the following value: password
client_id	The ID that Azure AD created when you registered the application.
scope	Use the following format for the value: `api://<organization ID>.<server URL>.<scope name>`
username	Azure AD user name.
password	Azure AD password.
client_secret	Client secret for the application.

A successful response includes the JWT access token in the access_token field. Copy the value and include it in a loginOAuth POST request to log in to Informatica Intelligent Cloud Services and start an API session.

The following text is an example of a response:


```
{
"token_type": "Bearer",
"scope": "api:// 38m52oDnrabtXYZb0TfwEf.dm1-us.informaticacloud.com/Custom.Read",
"expires_in": 4675,
"ext_expires_in": 4675,
"access_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6IjlHbW55RlBraGMzaE91UjIybXZTdmduTG83WSIsImtpZCI6IjlHbW55RlBraGMzaE91UjIybXZTdmduTG83WSJ9.eyJhdWQiOiJhcGk6Ly8zOG02SW9EbmthdGJYWUNiT1Rnd0VmLnFhLW1hLm1yZWwuaW5mYXFhLmNvbSIsImlzcyI6Imh0dHBzOi8vc3RzLndpbmRvd3MubmV0LzhmMmM1NDAzLTI3YTMtNDhiZC1hOWRhLTkyMjFjNDQwY2VkMi8iLCJpYXQiOjE2OTg4ODE1NzEsIm5iZiI6MTY5ODg4MTU3MSwiZXhwIjoxNjk4ODg2NTQ3LCJhY3IiOiIxIiwiYWlvIjoiQVZRQXEvOFVBQUFBR3pQZ0lvQ3N3VGdXNUdQUzhYUGdXbTlYKzNuWGRObTJ1TlZmbkVYZS9zSWgyUUJqTCt2Ymd1L2YrenpZT3Brck55MG5Ub2dmcE9vb3NwajF5VGJJVHdXU1Rpd2VZMzlwUFc3cURSZWNMR1U9IiwiYW1yIjpbInB3ZCIsIm1mYSJdLCJhcHBpZCI6IjgzNDM5MjNkLWRhZjMtNGVhMy05ZTRmLTg0ZTE3NWU0MGNmMCIsImFwcGlkYWNyIjoiMSIsImlwYWRkciI6IjMyLjE0MC4xMDQuMjE0IiwibmFtZSI6IkF6dXJldGVzdCIsIm9pZCI6IjYzYzY3OWFiLTlkMjQtNDljNC1iN2MyLTljODBlNzUxYmI3MCIsInJoIjoiMC5BYjBBQTFRc2o2TW52VWlwMnBJaHhFRE8wajJTUTRQejJxTk9uay1FNFhYa0RQRExBQzQuIiwic2NwIjoiQ3VzdG9tLlJlYWQiLCJzdWIiOiJfQTBYcl9oTGFiclNBRUNBLVM2Zk5LbmN1RjcxOWx4cXhPVGJsU0x3ZTMwIiwidGlkIjoiOGYyYzU0MDMtMjdhMy00OGJkLWE5ZGEtOTIyMWM0NDBjZWQyIiwidW5pcXVlX25hbWUiOiJhenVyZXRlc3RAcGxhdGZvcm1xYW9yZy5vbm1pY3Jvc29mdC5jb20iLCJ1cG4iOiJhenVyZXRlc3RAcGxhdGZvcm1xYW9yZy5vbm1pY3Jvc29mdC5jb20iLCJ1dGkiOiI1Y3RJN2dEcEJrZUlBaFhaclEtY0FBIiwidmVyIjoiMS4wIn0.HjeHZX5HNL05Ar2707pbU8MqiBainbWONCgMsWsX11XdmY8o5kCAI8J0_GTpqz3sNm1mZ7gm6De-ZLoIAgwDpdOCeuIovRM8-A_pWQfQ7IpuDndDPJNHyI1f-hc4zCyQlyuyD5AR_YbxxsFZE4SopmMCb_karAbKU2m1RdMuEVTB_4pVLXPTAz-3zHxf1OvooYnkrtsHa-x4JgCICnir5diAckm70JjPjqJNh0ks2F0jkqhrUGehAn29hiwQeTfjKgq4C8jToUSUs66W75C7Y6ageDq2bs_PMbPzbFOknQ8LqIH1sFaRT_VH1PK_YnfCIHHQtYZ1hw02OUTVQ6-6YA",
"id_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6IjlHbW55RlBraGMzaE91UjIybXZTdmduTG83WSJ9.eyJhdWQiOiI4MzQzOTIzZC1kYWYzLTRlYTMtOWU0Zi04NGUxNzVlNDBjZjAiLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubGluZS5jb20vOGYyYzU0MDMtMjdhMy00OGJkLWE5ZGEtOTIyMWM0NDBjZWQyL3YyLjAiLCJpYXQiOjE2OTg4ODE1NzEsIm5iZiI6MTY5ODg4MTU3MSwiZXhwIjoxNjk4ODg1NDcxLCJhaW8iOiJBV1FBbS84VUFBQUFBQkRCejM3cDd2aXY3eDFnVWMyOFVYcDIwODNMVjFFRXFCdkhMTWZZOXd1cFdqL3NvTmhXanZuR2ZIcjk2bHhTQ216bnVjVitIS29jUjBLaFpEZGdkRXFiN0hyeTlNZ050cHJwd2xhWWdRSDZoaURsTmRZRGRQVmIrR29GemdPUyIsInJoIjoiMC5BYjBBQTFRc2o2TW52VWlwMnBJaHhFRE8wajJTUTRQejJxTk9uay1FNFhYa0RQRExBQzQuIiwic3ViIjoiX0EwWHJfaExhYnJTQUVDQS1TNmZOS25jdUY3MTlseHF4T1RibFNMd2UzMCIsInRpZCI6IjhmMmM1NDAzLTI3YTMtNDhiZC1hOWRhLTkyMjFjNDQwY2VkMiIsInV0aSI6IjVjdEk3Z0RwQmtlSUFoWFpyUS1jQUEiLCJ2ZXIiOiIyLjAifQ.Q6ZnuVt760Zp7-Ev6Yb3EOHxu0ZRWKCnglPv85lgEzE_gVcUOhlN3_iIRONR1Q3S_2DeOZcTV1Y59nXonIuUevSQYaGGh3XDez6zopYA6jOMo2
```

```
eGUNZ4YC4sNUyXaapJmplhc8Dj3rvIumnUZ8tR78qdrv7ypy5VF2mUKo7U8QFYDK0uSi6y-
jXxEcjIOaUadDIJNFfLdezhyvsbme137iF_N1dzrZZzscQ-a6e2SJZJn4ueffd9RXfp8Eo__4p0YlMEVd_-
QOJzL3fGi29eBtfhbG3nmM_qSO8tTb-2_MXjXFB7HQBMMv6v5sCkwwvTKit-tmL8YWCNcnKVFpJxYg"
}
```

Fetch tokens using an authorization code

If you use multi-factor authentication, users can fetch a token using an authorization code.

To get the authorization code, use the following URI in your browser:

```
https://login.microsoftonline.com/<tenant ID>/oauth2/authorize?client_id=<client ID>
```

Append the following information to the URI:

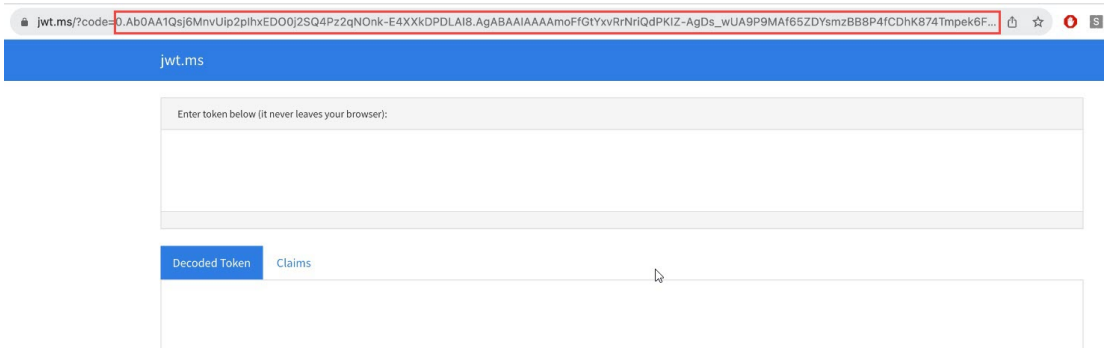
Field	Description
response_type	Use the following value: code
redirect_uri	The redirect URI that you set up for the application.
response_mode	Use the following value: query
scope	The scope that you defined for the application, followed by "openid"
state	Use the following value: 12345

For example:

```
https://login.microsoftonline.com/a412b1a7-bb12-4e62-a4d0-eff7034a60c6/oauth2/v2.0/authorize?
client_id=8173923d-daf4-2ea3-872f-84e176e40de0&response_type=code&redirect_uri=https://
jwt.ms&response_mode=query
&scope=api://8173923d-daf4-2ea3-872f-84e176e40de0/Custom.Read openid
&state=12345
```

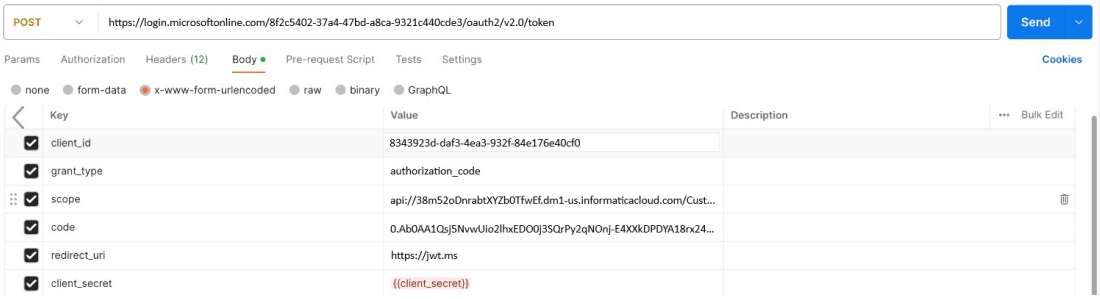
Press Enter. The authorization code appears in the URL.

Figure 4.



Send a POST request to get the token using the authorization code. For example:

Figure 5.



A successful response includes the JWT access token in the access_token field. Copy the value and include it in a loginOAuth POST request to log in to Informatica Intelligent Cloud Services and start an API session.

The following text is an example of a response:


```
{
"token_type": "Bearer",
"scope": "api:// 38m52oDnrabtXYZb0TfwEf.dm1-us.informaticacloud.com/Custom.Read",
"expires_in": 4675,
"ext_expires_in": 4675,
"access_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsIng1dCI6IjlHbW55RlBraGMzaE91UjIybXZTdmduTG83WSIsImtpZCI6Ijl
HbW55RlBraGMzaE91UjIybXZTdmduTG83WSJ9.eyJhdWQiOiJhcGk6Ly8zOG02SW9EbmthdGJYWUNiT1Rnd0VmLnFhLW1hLm
1yZWwuaW5mYXFhLmNvbSIsImlzcyI6Imh0dHBzOi8vc3RzLndpbmRvd3MubmV0LzhmMmM1NDAzLTI3YTMtNDhiZC1hOWRhLT
kyMjFjNDQwY2VkMi8iLCJpYXQiOjE2OTg4ODE1NzEsIm5iZiI6MTY5ODg4MTU3MSwiZXhwIjoxNjk4ODg2NTQ3LCJhY3IiOi
IxIiwiYWlvIjoiQVZRQXEvOFVBQUFBR3pQZ0lvQ3N3VGdXNUdQUzhYUGdXbTlYKzNuWGRObTJ1TlZmbkVYZS9zSWgyUUJqTC
t2Ymd1L2YrenpZT3Brck55MG5Ub2dmcE9vb3NwajF5VGJJVHdXU1Rpd2VZMzlwUFc3cURSZWNMR1U9IiwiYW1yIjpbInB3ZC
IsIm1mYSJdLCJhcHBpZCI6IjgzNDM5MjNkLWRhZjMtNGVhMy05ZTRmLTg0ZTE3NWU0MGNmMCIsImFwcGlkYWNyIjoiMSIsIm
lwYWRkciI6IjMyLjE0MC4xMDQuMjE0IiwibmFtZSI6IkF6dXJldGVzdCIsIm9pZCI6IjYzYzY3OWFiLTlkMjQtNDljNC1iN2
MyLTljODBlNzUxYmI3MCIsInJoIjoiMC5BYjBBQTFRc2o2TW52VWlwMnBJaHhFRE8wajJTUTRQejJxTk9uay1FNFhYa0RQRE
xBQzQuIiwic2NwIjoiQ3VzdG9tLlJlYWQiLCJzdWIiOiJfQTBYcl9oTGFiclNBRUNBLVM2Zk5LbmN1RjcxOWx4cXhPVGJsU0
x3ZTMwIiwidGlkIjoiOGYyYzU0MDMtMjdhMy00OGJkLWE5ZGEtOTIyMWM0NDBjZWQyIiwidW5pcXVlX25hbWUiOiJhenVyZX
Rlc3RAcGxhdGZvcm1xYW9yZy5vbm1pY3Jvc29mdC5jb20iLCJ1cG4iOiJhenVyZXRlc3RAcGxhdGZvcm1xYW9yZy5vbm1pY3
Jvc29mdC5jb20iLCJ1dGkiOiI1Y3RJN2dEcEJrZUlBaFhaclEtY0FBIiwidmVyIjoiMS4wIn0.HjeHZX5HNL05Ar2707pbU8
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A_pWQfQ7IpuDndDPJNHyI1f-
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x4JgCICnir5diAckm70JjPjqJNh0ks2FOjkqhrUGehAn29hiwQeTfjKgq4C8jToUSUs66W75C7Y6ageDq2bs_PMbPzbFOknQ
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"id_token":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImtpZCI6IjlHbW55RlBraGMzaE91UjIybXZTdmduTG83WSJ9.eyJhdWQiOi
I4MzQzOTIzZC1kYWYzLTRlYTMtOWU0Zi04NGUxNzVlNDBjZjAiLCJpc3MiOiJodHRwczovL2xvZ2luLm1pY3Jvc29mdG9ubG
luZS5jb20vOGYyYzU0MDMtMjdhMy00OGJkLWE5ZGEtOTIyMWM0NDBjZWQyL3YyLjAiLCJpYXQiOjE2OTg4ODE1NzEsIm5iZi
I6MTY5ODg4MTU3MSwiZXhwIjoxNjk4ODg1NDcxLCJhaW8iOiJBV1FBbS84VUFBQUFBQkRCejM3cDd2aXY3eDFnVWMyOFVYcD
IwODNMVjFFRXFCdkhMTWZZOXd1cFdqL3NvTmhXanZuR2ZIcjk2bHhTQ216bnVjVitIS29jUjBLaFpEZGdkRXFiN0hyeTlNZ0
50cHJwd2xhWWdRSDZoaURsTmRZRGRQVmIrR29GemdPUyIsInJoIjoiMC5BYjBBQTFRc2o2TW52VWlwMnBJaHhFRE8wajJTUT
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Ev6Yb3EOHxu0ZRWKCnglPv85lgEzE_gVcUOhlN3_iIRONR1Q3S_2DeOZcTV1Y59nXonIuUevSQYaGGh3XDez6zopYA6jOMo2
eGUNZ4YC4sNUyXaapJmplhc8Dj3rvIumnUZ8tR78qdrv7ypy5VF2mUKo7U8QFYDK0uSi6y-
jXxEcjIOaUadDIJNFfLdezhyvsbme137iF_N1dzrZZzscQ-a6e2SJZJn4ueffd9RXfp8Eo__4p0YlMEVd_-
QOJzL3fGi29eBtfhbG3nmM_qSO8tTb-2_MXjXFB7HQBMMv6v5sCkwwvTKit-tmL8YWCNcnKVFpJxYg"
}
```


Author

Informatica Cloud Trust Team